

Youngjin Moon

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291, Daehak-ro, Yuseong-gu, Daejeon, Republic of Korea

SUMMARY

• Expertise in AI/ML Acceleration ASIC Design

- Digital Deep Learning Accelerator
- Hardware-Software Co-design (Skilled in sparsity optimization and model compression)

• End-to-end IC Implementation & Verification

- Multiple Tape-outs including Video AI accelerator and Large Language Model accelerator
- Full stack digital (RTL, Synthesis, PnR) chip design experiences

EDUCATION

• Ph.D Candidate in AI Semiconductor, KAIST

Mar. 2026 -

- Advisor: Hoi-Jun Yoo, KAIST (IEEE Fellow since 2008)

Daejeon, Republic of Korea

• M.S in AI Semiconductor, KAIST

Mar. 2024 - Feb. 2026

- Advisor: Hoi-Jun Yoo, KAIST (IEEE Fellow since 2008)

Daejeon, Republic of Korea

- Thesis: An Energy-Efficient Real-Time Mamba-Based Video Understanding Accelerator for Edge Devices

• B.S in Mechanical Engineering & Electrical Engineering, SKKU

Mar. 2017 - Feb. 2024

- GPA: 4.03/4.50

Suwon, Republic of Korea

- Major1(2) GPA: 4.01(4.05)/4.50

WORK EXPERIENCE

• Samsung Electronics

Aug. 2022 - Dec. 2022

Long-term Internship in Modem Development Team, System LSI

Suwon, Republic of Korea

- Analyze High Speed Digital Interface Architecture
- Design Sub-module of High Speed Digital Interface

• Republic of Korea Army

Sep. 2018 - Apr. 2020

Mandatory Military Service

Cheonan, Republic of Korea

- Trained, disciplined, and instructed soldiers in various military skills
- Gained leadership and operational experience as a Duty Officer

SKILLS

- **ASIC Design:** Digital Processor & Deep Learning Accelerator Design, Hardware-Software Co-design
- **Programming Languages:** Verilog HDL, Python, C/C++
- **EDA Tools:** Synopsys Design Compiler, IC Compiler II, PrimeTime, Cadence Virtuoso
- **Language:** English (fluent), Korean (native)

REPRESENTATIVE PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PATENT, S=IN SUBMISSION, T=THESIS

- [C.1] [S.VLSI 2026] **Youngjin Moon**, Sangwoo Ha, Junha Ryu, Jungjun Oh, Jingu Lee, Wooyoung Jo, Minsung Kim, and Hoi-Jun Yoo. "NOVA: A 5.3mJ/frame Fully Integrated Neural Video SoC with Content-Adaptive Architecture for 4K Real-time Encoding/Decoding and Processing", 2026 IEEE Symposium on VLSI Technology and Circuits (S.VLSI)
- [C.2] [ISCAS 2025] **Youngjin Moon**, Sangwoo Ha, Soyeon Kim, Junha Ryu, Hoi-Jun Yoo, and Donghyeon Han. "A 2.67 mJ/frame Video Mamba Accelerator with Importance-aware Redundancy Elimination and SSM Computing Reformulation", 2025 IEEE International Symposium on Circuits and Systems (ISCAS)
- [C.3] [ASSCC 2026] Sangmyoung Lee, Sangwoo Ha, **Youngjin Moon**, Minsung Kim, Seryeong Kim, Wooyoung Jo, Junhyuk Lee, Sangyeob Kim, and Hoi-Jun Yoo. "NeuLPU: A 3.6 mW Spiking LLM Processing Unit for On-device Long-context Inference", 2026 IEEE Asian Solid-State Circuits Conference (ASSCC)

- [C.4] [ASSCC 2026] Jungwan Lee, Seongyon Hong, **Youngjin Moon**, Jingu Lee, Junhyuk Lee, Yuseon Choi, Jungjun Oh, and Hoi-Jun Yoo. "VectorFLASH: a 2.8-8.7 Tokens/S Resource-Efficient Long-Context LLM Accelerator with Dual Vector Quantization on Edge FPGA ", 2026 IEEE Asian Solid-State Circuits Conference (ASSCC)
- [C.5] [MICRO 2026] Sangwoo Ha, Hyunwoo Seo, Yurim Jo, **Youngjin Moon**, and Hoi-Jun Yoo. "EdgeXpert: An Edge Device for Memory-Efficient LLM Inference with Mixture-of-Experts and Speculative Decoding", 2026 IEEE/ACM International Symposium on Microarchitecture (MICRO)
- [C.6] [COOLCHIPS 2026] Sangwoo Ha, Jingu Lee, **Youngjin Moon**, Sunjoo Whang, Wooyoung Jo, Gwangtae Park, Soyeon Um, Junha Ryu, Yurim Jo, Jongjun Park, and Hoi-Jun Yoo. "An Energy-Efficient LLM Processor for Mobile Devices with Pipelined NPU-CIM", 2026 IEEE Symposium on Low-Power and High-Speed Chips (COOL Chips)
- [C.7] [ISSCC 2026] Sangwoo Ha, Jingu Lee, **Youngjin Moon**, Sunjoo Whang, Wooyoung Jo, Gwangtae Park, Soyeon Um, Junha Ryu, Yurim Jo, and Hoi-Jun Yoo. "SMoLPU: A 122.1 μ J/token Sparse MoE-based Speculative Decoding LLM Processing Unit with Adaptive-Offload NPU-CIM Core", 2026 IEEE International Conference on Solid-State Circuits (ISSCC)
- [C.8] [S.VLSI 2025] Soyeon Kim, Hankyul Kwon, Jingu Lee, **Youngjin Moon**, Hongseok Lee, Junha Ryu, Zhamaliddin Kalzhan, Sangyeob Kim, Wooyoung Jo, and Hoi-Jun Yoo. "NuVPU: A 4.8 9.6 mJ/frame Progressive NTT-based Unified Video Processor for Stable Video Streaming and Processing with Neural Video Codec", 2025 IEEE Symposium on VLSI Technology and Circuits (S.VLSI)
- [J.1] [TCSVT 2026] **Youngjin Moon**, Sangwoo Ha, Junha Ryu, Soyeon Kim, Donghyeon Han, and Hoi-Jun Yoo. "An Energy-Efficient State-Space Model Accelerator for Real-Time Video Understanding via Redundancy-Conscious Design", IEEE Transactions on Circuits and Systems for Video Technology, June. 2026
- [J.2] [TCAS-II 2025] Sangmyoung Lee, Seryeong Kim, Jongjun Park, **Youngjin Moon**, Minsung Kim, Junha Ryu, Hoi-Jun Yoo, Donghyeon Han. "A 227.1 TOPS/W High Energy-efficiency PNN-based 3D Object Recognition Processor with Spiking Neural Network for Edge Device", IEEE Transactions on Circuits and Systems II, Jul. 2025

HONORS AND AWARDS

- **Academic Excellent Scholarship**
Sungkyunkwan University, 2017-1st semester GPA 4.5/4.5 Sep. 2017
- **Global Student Success Creative Scholarship**
Sungkyunkwan University Sep. 2021
- **S-Hero Gold Prize**
Ministry of Science and ICT of Korea, Field Customized Science and Engineering Project Sep. 2021
- **Intelligent Connect Leading University Competition Grand Prize**
Sungkyunkwan University, Creative Research Presentation Competition Oct. 2021
- **X-Corps Festival Grand Prize (Award of Ministry of Science and ICT)**
Ministry of Science and ICT of Korea, Undergraduate Research Project Contest Nov. 2021
- **SAMSUNG Dream Class Mentor Scholarship**
SAMSUNG, Mentoring middle school students Nov. 2022
- **IEEE ISCAS Student Travel Grant Award**
IEEE Circuits and Systems Society (CASS) Apr. 2025
- **IITP President's Award**
Institute for Information & communication Technology Planning & evaluation(IITP) of Korea, AI Semiconductor Contest Jun. 2025

REFERENCES

1. **Hoi-jun Yoo**
IEEE Fellow, Professor, KAIST
Email: hjyoo@kaist.ac.kr
Thesis Advisor